

TouchBistro Challenge

CxC Datathon

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Introduction

This report details the steps taken for developing a lightweight and efficient global model from the TouchBistro dataset to determine key insights that determine sales revenue. Detailed are the steps from key insights taken from the data, preprocessing, model building, and visualization.

Our goal and mission was to develop a model and provide key insights for venues and businesses. We decided to focus our attention at providing actionable insights towards individual businesses.

Key Insights

Here are some key insights determined from observations in the dataset:

1. There are 24 features from the bills and venues datasets
2. Nearly 9 million data points present in the bills dataset. We have decided to go ahead and only use 100,000 to develop our model. This decision was made to reflect real world scenarios for small, individual businesses. Ensuring that any restaurant can follow our methodologies to develop their own accurate model tailored towards their needs
3. Out of all of the features, only concepts contained missing values

Preprocessing

Present in this section are the steps we took, the rationale behind why each step was taken and key insights determined through each.

1. Joining the datasets: The bills.csv and venues.csv were joined through their common column of venue_xref_id.
 - a. The decision to combine this was to make preprocessing in further steps easier, separating these datasets serves no notable purpose and only means to complicate the next few steps
2. Dropping unneeded features: Uuid columns such as waiter_uuid, bill_uuid, and order_uuid were dropped. However, the venue_xref_id column was left intact.
 - a. The reason for dropping the uuid columns is that we want to generalize for businesses rather than each individual order or waiter, hence venue_xref_id was left while the uuid columns were dropped.
3. Handling missing values: As stated previously, the concept was the only column that contained missing values, so we decided to use the KNN Imputer technique to fill in these missing values.
 - a. Since we were intentionally working with a small dataset the option to drop missing rows was not a feasible option.
 - b. Looking through the unique values of the concept, it has a relatively low cardinality, meaning the data could be encoded and KNN clustering could be used as an imputer.
 - c. The steps taken to impute required us to determine feature columns that we wish to use to advise the imputer. The decision was made that likely the most important features were the sales revenue, the gratuity, discount items, and total billed. This was done to highlight the main differences between different types of venues, for example, gratuity is an important feature for venues such as bars and family dining.
4. Object encoding: Object columns need to be converted to numerical values for machine learning.
 - a. 2 types of objects were observed, strings and dates
 - b. To handle strings, one hot encoding was used for those with low cardinality. The reason to use one hot encoding rather than ordinal encoding is because ordinal encoding suggests a particular order, while one hot doesn't, however is not suitable for large cardinality. None of the categorical features have any apparent ordering. Target encoding was used for those with high cardinality, this is a standard technique to handle such cases.
 - c. To handle datetime, a more individualized approach was taken. The business date feature was first converted to unix time and normalized, this is to ensure that the data is properly standardized for the model and to capture temporal information. We want to know if a datapoint happened after, but don't want all of them to be skewed as high values if normalization wasn't done to the unix time. The order time features, such as when an order happened, duration of order, and when the order was completed were reworked.

- d. The end of the order is redundant when having the duration and the start, so that column was removed. Having the full time of the order also is redundant since we know what day it occurred due to the business date, so that feature was transformed to be one hot encoded to represent the hour it occurred. This is because we don't want to give a specific time precedence since bars operate during the night while family dining restaurants might close early. The duration of the order was transformed to a float representing the minutes.
- 5. Additional and removal of features: Additional features were added to provide more context to the model as well as provide more insights depending on what the model found had the highest correlation. Some other features were also removed due to redundancy and out of necessity
 - a. Seasonality was added, information about the model which season the order occurred at.
 - b. The day of the week was added.
 - c. Tip percentage was added, a ratio of the gratuity of the bill percentage
 - d. Features that revealed data about actual sales revenue were removed like bill total billed, as well as information about the tax paid since that has a linear correlation with the revenue. Gratuity amount was removed due to providing information in combination with tip percentage of the actual value. These features were removed because in a real world scenario, this information would not have been present before the sales revenue.

At the end, our model is to predict the sales revenue with only information of the city, concept, time of day, day of the week, duration of the order, and tip percentage, providing for a general purpose model.

Model Runtime:

Instead of trying to predict the sales revenue exactly, we will be trying to predict a relative value. The issue with trying to predict the sales revenue is that it differs venue to venue. Even if two venues are of the same concept in the same city, they could be vastly different sizes, one could be generating \$5000 of daily sales while another generate \$10,000+.

We will perform a scaler of the revenue and our model will provide a scaled value using the sklearn's standard scaler. This will mean that the model can generalize and provide a value that let's businesses know whether the current day will yield more or less sales relative to the venue.

Here are the steps taken to ensure this approach:

1. We separate the dataset based on each venue
2. We drop any venues with less than 20 rows to remove outliers and not enough data
3. We will perform standard scaler on each venue data separately and store each venue's scaler in a dictionary
4. During the data splitting, we will make sure that at least each unique venue entry is present in the test/validation dataset.
5. During training, these scaled data will be used
6. During testing and validation, the output of the model will be inversely scaled based on the test venue found from the dictionary. This will scale up the output to match the current test's, letting us know how accurate it is.

As for models, we tried an LSTM model at the start, but it gave terrible results, and frankly we believed classical models would perform better.

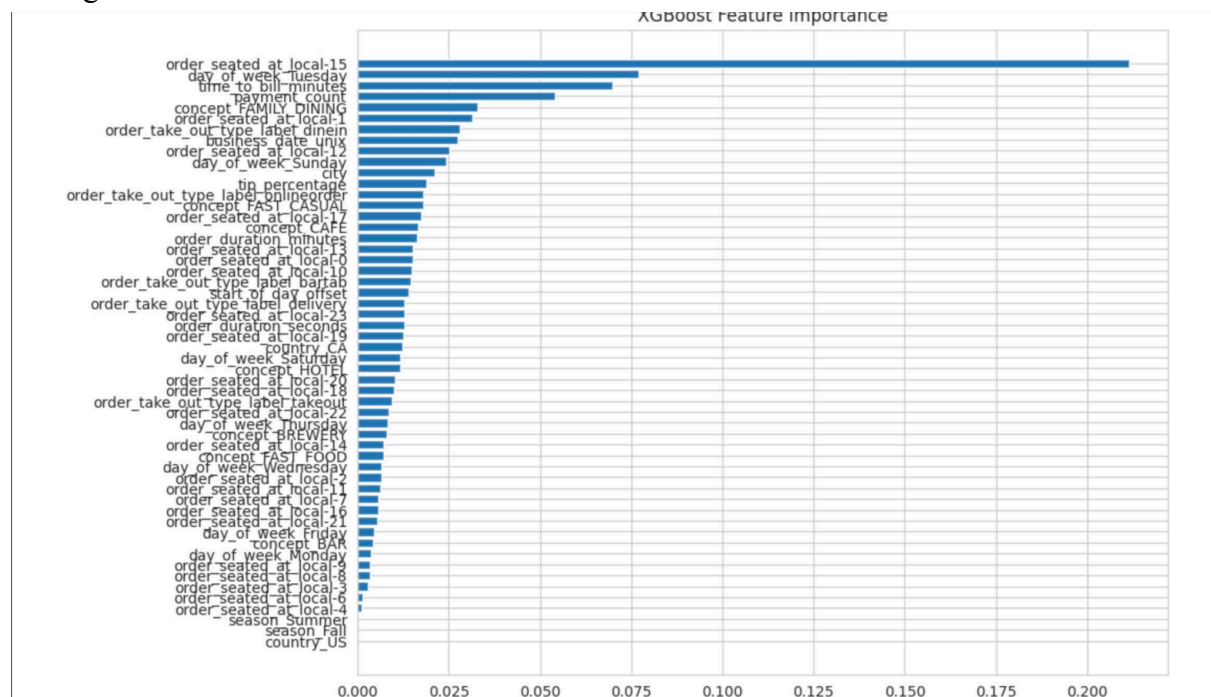
The main model used was XGBoost for its high accuracy and moderate complexity. An XGBoost model was much more lightweight than a deep learning model, while also grasping much of the complexity that too simplistic classical models like linear regression cannot.

Grid search was used to test our various hyperparameters along with group kfold for cross validation.

Results

The model was able to achieve an rsme value of 0.436. These are great results for a limited dataset, a high variability dataset, and being non-deterministic with the given dataset (sales revenue can be completely different even with the exact same features due to other factors such as market variability, government policies, and other outside factors not present in the dataset).

We had achieved our goal of developing an efficient and high accuracy model for the given dataset, this also allowed us to have confidence in feature extraction from the model and driving conclusive results:



Key insights:

1. An order occurring at 3pm was the biggest contributing feature to the model accuracy, meaning that 3pm is very consistent with the sales revenue, other times perhaps showing high variability while 3pm seems to be a good driver of traffic
2. Tuesday's also provide good identification for predicting sales revenue, this doesn't necessarily mean that Tuesday's are the busiest, but rather that Tuesday's provide consistent variability.
3. The time of the bill was one of the highest predictors, this suggests that the longer the order the higher sales revenue
4. Seasonality was a low predictor, meaning that in general the season doesn't have a strong correlation with sales revenue.